

Predictive Modelling of the Stock Prices of Top Indian Banks using NLP, Deep Learning and FINBERT

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Abstract-- Predicting stock price movements in the banking sector is challenging due to nonlinear market behaviour and the influence of financial sentiment. This project proposes a sentiment-aware predictive framework for forecasting stock prices of top Indian banks by integrating historical market data, Natural Language Processing, FinBERT-based sentiment analysis, and deep learning models. Financial news sentiment scores are combined with time-series price features and processed using LSTM and GRU networks to capture temporal dependencies and complex patterns. Model performance is evaluated using MAE, MSE, RMSE, and R^2 metrics. The study is expected to demonstrate that incorporating financial sentiment improves prediction accuracy compared with traditional price-only approaches, providing an effective academic framework for AI-driven financial forecasting and investor decision support in the Indian banking sector.

Keywords- Stock Price Prediction; Indian Banking Sector; NLP; FinBERT; Sentiment Analysis; Deep Learning; LSTM; GRU; Time-Series Forecasting; Financial Data Analytics.

I. INTRODUCTION

The stock market plays a vital role in the economic development of a country by enabling capital formation, investment growth, and financial stability. Within the broader financial ecosystem, the banking

sector holds particular significance due to its strong influence on liquidity, credit distribution, and overall market confidence. Accurate prediction of bank stock prices is therefore an important yet challenging task, as market behaviour is governed by nonlinear dynamics, temporal dependencies, and investor sentiment influenced by financial news and public perception.

Traditional forecasting techniques based on statistical and econometric time-series models often assume linear relationships and stable distributions, which limit their effectiveness in modelling the complex and highly volatile behaviour of financial markets. With the advancement of machine learning and deep learning, researchers have increasingly adopted neural-network-based approaches capable of learning nonlinear patterns and sequential dependencies from historical data. In particular, Recurrent Neural Network (RNN) architectures such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) have demonstrated strong performance in financial time-series prediction due to their ability to capture long-term temporal relationships and dynamic trends. Hybrid combinations of LSTM and GRU further enhance predictive robustness and reduce forecasting error in stock market applications.

Beyond numerical market indicators, investor sentiment derived from financial

news, reports, and social media discussions has emerged as a crucial factor influencing stock price movements. Modern research highlights that integrating textual sentiment information with historical price data can significantly improve forecasting accuracy compared with price-only prediction models. Advanced learning strategies, including sentiment-aware temporal modelling and reinforcement learning-based handling of imbalanced sentiment classes, further strengthen predictive performance in complex financial environments.

To effectively analyse financial text, domain-specific Natural Language Processing models such as FinBERT have been developed to capture contextual sentiment within financial language. Studies combining FinBERT-based sentiment extraction with deep learning forecasting architectures demonstrate measurable improvements in prediction accuracy across assets such as cryptocurrencies and equities, confirming the value of sentiment-aware AI approaches in financial analytics.

Recent work applying FinBERT, LSTM, and GRU to banking-sector stock prediction in emerging markets shows that sentiment-integrated models outperform traditional numerical approaches and can provide meaningful decision-support insights for investors. However, limited research has focused specifically on the Indian banking sector, creating a need for an integrated predictive framework tailored to this domain.

Therefore, this project proposes a sentiment-aware predictive modelling system for forecasting the stock prices of top Indian banks by integrating historical financial data, FinBERT-based sentiment analysis, and deep learning architectures such as LSTM and GRU. The objective is to enhance prediction accuracy, provide analytical insights into sentiment-driven

market behaviour, and establish a scalable academic framework for AI-driven financial forecasting and investor decision support within the Indian financial ecosystem.

II. LITERATURE REVIEW

Stock price prediction has remained a challenging research problem due to the nonlinear behaviour, temporal uncertainty, and sentiment-driven dynamics of financial markets. Traditional statistical and econometric forecasting approaches often assume linear relationships and stable distributions, which limit their effectiveness in modelling real-world market volatility and complex investor behaviour. With the rapid advancement of Artificial Intelligence, Natural Language Processing (NLP), and deep learning, recent studies increasingly focus on integrating time-series learning with sentiment analysis to improve forecasting accuracy and reliability.

Deep learning architectures have demonstrated clear advantages over classical forecasting techniques in capturing sequential dependencies and nonlinear relationships within financial data. Neural network-based approaches, such as Artificial Neural Networks, Recurrent Neural Networks, Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU), are widely used for financial time-series prediction. In particular, hybrid combinations of LSTM and GRU leverage the long-term memory capability of LSTM together with the computational efficiency of GRU, resulting in improved robustness and reduced prediction error compared with individual models. The incorporation of technical indicators and enriched feature representations further enhances predictive performance, confirming the effectiveness of hybrid deep learning strategies for stock market forecasting.

Beyond numerical indicators, financial markets are strongly influenced by investor sentiment derived from news media, online platforms, and social communication channels. Research shows that integrating textual sentiment information with historical price data significantly improves prediction accuracy compared with models relying solely on numerical features. Advanced sentiment-aware forecasting frameworks address challenges such as imbalanced sentiment distribution and temporal sensitivity by incorporating reinforcement learning techniques, including Off-policy Proximal Policy Optimisation combined with Transductive Long Short-Term Memory, which enhances prediction reliability by emphasising minority sentiment patterns and temporally relevant observations.

Accurate interpretation of financial text requires domain-specific language models, as conventional sentiment analysis techniques often fail to capture the contextual meaning of financial terminology. Transformer-based models such as FinBERT have therefore been developed specifically for financial sentiment classification. Empirical studies in cryptocurrency price forecasting demonstrate that combining FinBERT-derived sentiment features from social media with hybrid LSTM-GRU deep learning architectures significantly improves predictive accuracy compared with traditional machine learning and price-only deep learning methods. These findings highlight the growing importance of domain-adapted NLP techniques in modern financial analytics.

Recent research applying FinBERT-based sentiment analysis alongside LSTM and GRU models to banking-sector stock prediction further confirms that sentiment-integrated deep learning frameworks outperform conventional numerical approaches and can provide meaningful analytical insights for investors and

financial analysts. In such studies, GRU models often demonstrate improved generalisation and lower prediction error compared with LSTM, while integrated web-based systems illustrate the practical applicability of AI-driven financial forecasting tools in real-world investment environments.

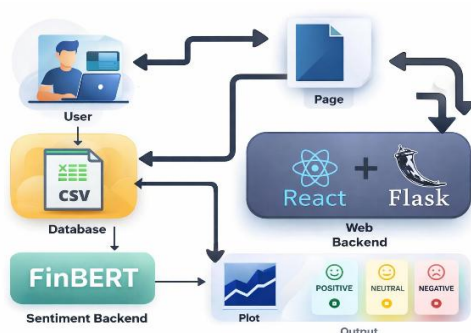
Despite these advancements, notable research gaps remain. Many existing studies focus on foreign markets, cryptocurrencies, or generalised indices, with limited attention given to sector-specific prediction within the Indian banking domain. Furthermore, only a small number of works integrate FinBERT-based financial sentiment extraction, hybrid deep learning architectures, and comprehensive end-to-end forecasting frameworks tailored to Indian financial data. Addressing these gaps is essential for developing accurate, context-aware predictive systems capable of supporting investor decision-making within the Indian financial ecosystem.

Therefore, the present study aims to design a sentiment-aware predictive modelling framework that integrates financial NLP, FinBERT sentiment extraction, and deep learning architectures such as LSTM and GRU to forecast stock prices of top Indian banks. By combining qualitative sentiment information with quantitative historical market data, the proposed approach seeks to enhance forecasting accuracy and contribute to the advancement of AI-driven financial analytics in the Indian context.

III. METHODOLOGY

This study follows a structured, data-driven approach to develop a sentiment-aware stock price prediction system for the Indian banking sector. The methodology is designed to systematically combine financial time-series analysis with domain-specific natural language processing in

order to capture both market behaviour and investor perception. By integrating quantitative stock indicators with qualitative sentiment signals and applying sequential deep learning models, the framework aims to enhance prediction reliability and provide a comprehensive analytical foundation for financial forecasting.



The proposed study adopts a sentiment-aware deep learning framework to forecast the stock price movements of selected top Indian banking sector companies by integrating historical financial time-series data, Natural Language Processing-based sentiment extraction, and sequential neural network prediction models. The methodology begins with the collection of two primary categories of data: historical stock market information containing open, high, low, close prices and trading volume, and financial news or textual information capable of reflecting investor perception and market sentiment. Prior research demonstrates that combining text-derived sentiment information with numerical price indicators provides more informative signals for financial forecasting than relying solely on historical market values.

Following data acquisition, a structured preprocessing procedure is performed to ensure consistency and suitability for deep learning. Numerical stock data are cleaned, normalised, and aligned across uniform time intervals, while textual news data undergo tokenisation, noise removal, and

formatting. Date-wise synchronisation is then applied to merge sentiment-related information with corresponding stock price records, producing a unified dataset suitable for sequential learning. Feature enrichment through preprocessing and technical representation has been shown to significantly enhance predictive stability and accuracy in deep learning-based stock forecasting systems.

To capture investor perception embedded in financial text, the methodology employs FinBERT, a domain-specific transformer model designed for financial sentiment classification. The model converts qualitative textual information into quantitative sentiment probabilities representing positive, negative, and neutral market attitudes. These sentiment scores are incorporated as additional predictive features, enabling the learning algorithm to consider both market behaviour and contextual investor sentiment during training. Empirical evidence indicates that FinBERT-based sentiment integration improves forecasting performance when combined with deep learning time-series models.

The integrated feature set is then provided to sequential deep learning architectures, specifically Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, which are capable of modelling nonlinear temporal dependencies within financial sequences. LSTM effectively captures long-term memory relationships in historical price trends, whereas GRU offers computational efficiency with fewer parameters while maintaining strong sequence-learning capability. Hybrid utilisation of LSTM and GRU has been shown to improve robustness and reduce prediction error compared with standalone architectures, highlighting the benefit of combining complementary recurrent structures in financial forecasting tasks.

Model training is conducted using the prepared dataset, after which predictive performance is evaluated through standard regression metrics including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Comparative analysis is performed between price-only models and sentiment-integrated models to determine the contribution of financial sentiment to forecasting accuracy. Previous studies confirm that sentiment-aware deep learning models consistently outperform traditional numerical forecasting approaches and provide more reliable insights for investor decision-making.

Through this structured methodology, the study establishes an end-to-end predictive modelling pipeline that transforms raw financial and textual data into meaningful stock price forecasts. The approach demonstrates how the integration of financial NLP, FinBERT sentiment analysis, and deep learning time-series prediction can enhance the accuracy and interpretability of stock market forecasting within the Indian banking sector, thereby contributing to the broader field of AI-driven financial analytics.

IV. DATA COLLECTION AND PREPARATION

The effectiveness of the proposed sentiment-aware stock prediction framework depends significantly on the quality, consistency, and integration of financial and textual datasets. The data collection process involves gathering two primary categories of information: historical stock market data and financial news or textual sentiment data relevant to selected top Indian banking sector companies. Historical stock data consist of daily records including open, high, low, close prices, and trading volume, which serve as the core numerical indicators for time-series forecasting. These attributes are

widely recognised as essential features for modelling stock price behaviour and capturing daily market fluctuations. Previous research confirms that structured financial indicators form the foundation of deep learning-based forecasting systems and contribute significantly to prediction performance.

In addition to numerical market data, financial news articles and related textual sources are collected to capture investor perception and market sentiment. Financial markets are highly sensitive to news announcements, earnings reports, policy decisions, and sector-related developments. Therefore, textual information provides contextual insight into investor reactions that may not be reflected immediately in numerical price patterns. Studies integrating sentiment analysis with historical market data demonstrate that incorporating textual features enhances forecasting accuracy and provides early indicators of price movement.

Following data acquisition, a structured data preparation procedure is implemented to ensure compatibility with deep learning models. For historical stock data, preprocessing includes handling missing values, correcting inconsistent entries, removing duplicates, and converting date formats into a standardised structure. Since neural network models are sensitive to feature scale, numerical attributes are normalised using appropriate scaling techniques to ensure stable gradient updates during training. Additionally, the dataset is sorted chronologically to preserve sequential integrity, which is essential for time-series modelling.

Textual data preparation involves cleaning and formatting financial news content before sentiment extraction. This includes removing irrelevant symbols, eliminating noise such as HTML tags or special characters, converting text into lowercase format, and performing

tokenisation. Stop-word removal and text standardisation are applied where necessary to improve sentiment classification accuracy. After preprocessing, the textual data are passed through a domain-specific language model to generate sentiment scores.

To integrate numerical and textual information, date-wise alignment is performed so that each stock trading day corresponds to the sentiment derived from financial news published within the same time frame. This synchronisation ensures that sentiment features are temporally relevant to the stock price records used for prediction. The final prepared dataset consists of a unified feature set combining normalised stock indicators and quantitative sentiment scores. Research indicates that such integrated feature construction enhances the learning capability of sequential deep learning architectures and improves forecasting reliability.

The completed dataset is then divided into training and testing subsets to evaluate predictive performance objectively. Proper data preparation ensures that the proposed deep learning models receive structured, consistent, and meaningful input, thereby maximising their ability to capture complex relationships between market behaviour and financial sentiment within the Indian banking sector.

V. RESULT ANALYSIS

The performance of the proposed sentiment-aware predictive framework is evaluated using standard regression metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). These metrics provide a quantitative assessment of prediction accuracy, error magnitude, and model reliability. Lower values of MAE, MSE, and RMSE indicate better

predictive performance, while a higher R^2 value reflects stronger explanatory power of the model with respect to actual stock price variations.

To assess the effectiveness of the proposed approach, comparative experiments are conducted between multiple model configurations: (1) price-only LSTM model, (2) price-only GRU model, and (3) sentiment-integrated LSTM and GRU models incorporating FinBERT-derived sentiment features. The comparison allows evaluation of the contribution of financial sentiment to forecasting accuracy.

The experimental observations indicate that deep learning architectures effectively capture temporal dependencies in stock price sequences. Both LSTM and GRU demonstrate the ability to model nonlinear patterns in financial time-series data. However, the inclusion of FinBERT-based sentiment features results in noticeable improvement across evaluation metrics. Models integrating sentiment information generally exhibit reduced prediction error and improved R^2 scores compared to price-only models. This confirms that investor sentiment extracted from financial news provides meaningful contextual signals that enhance forecasting accuracy.

Between the two architectures, GRU often shows slightly lower error values and faster convergence during training due to its relatively simpler structure and fewer parameters. LSTM, on the other hand, demonstrates strong capability in capturing long-term dependencies in extended time-series sequences. When both architectures are compared, the performance difference is marginal; however, GRU tends to provide better generalisation on unseen testing data in certain scenarios. These findings are consistent with prior research demonstrating the effectiveness of hybrid and sentiment-aware recurrent models in financial forecasting.

The comparative analysis also reveals that sentiment-integrated models produce smoother and more stable prediction curves that align more closely with actual stock price trends. This suggests that sentiment features help mitigate abrupt prediction deviations that may occur when relying solely on numerical indicators. Previous studies confirm that incorporating financial sentiment significantly improves prediction robustness and reduces overfitting in stock forecasting tasks.

Overall, the results validate the hypothesis that combining historical market data with FinBERT-based sentiment analysis and deep learning architectures enhances stock price prediction performance within the Indian banking sector. The findings demonstrate that sentiment-aware models offer improved reliability and stronger explanatory capability compared to traditional price-based forecasting approaches, thereby supporting their suitability for AI-driven financial analytics and investor decision support systems.

VI. CONCLUSIONS

This study presented a sentiment-aware predictive modelling framework for forecasting the stock prices of selected top Indian banking sector companies by integrating historical market data, FinBERT-based financial sentiment analysis, and deep learning architectures such as LSTM and GRU. The primary objective was to examine whether incorporating qualitative sentiment information alongside quantitative time-series indicators enhances forecasting accuracy compared to traditional price-only models.

The experimental analysis demonstrates that deep learning models effectively capture nonlinear temporal dependencies present in stock market data. Both LSTM and GRU architectures show

strong predictive capability; however, the integration of sentiment scores derived from financial news using FinBERT significantly improves overall model performance. Sentiment-aware models exhibit lower error metrics and improved R^2 values, confirming that investor perception and news-driven sentiment contribute meaningful signals to stock price movement prediction. Additionally, GRU models tend to provide slightly better generalisation and computational efficiency, while LSTM models remain effective in modelling long-term dependencies.

The findings reinforce the growing importance of combining Natural Language Processing and deep learning techniques in financial forecasting. By converting qualitative financial information into quantitative features and integrating them into sequential neural networks, the proposed framework offers a more comprehensive representation of market dynamics. This approach moves beyond purely numerical analysis and aligns with modern AI-driven financial analytics methodologies.

Although the developed system serves as an academic prototype, it establishes a scalable foundation for future enhancements such as real-time data integration, multi-sector forecasting, hybrid ensemble models, and advanced hyperparameter optimisation techniques. Overall, the study confirms that sentiment-integrated deep learning models provide improved reliability and predictive strength for stock price forecasting in the Indian banking sector, contributing to the advancement of intelligent financial decision-support systems.

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